

	$6.62 \times 10^{-25} = 8.36 \times 10^{-25} \sin \alpha$ $\alpha = 52.4^\circ$
6 (a) (i)	Acceleration is the rate of change of velocity.
(ii)	At locations near the surface of the Earth, the distance between vertical positions is small compared to the radius of the Earth, R_E. Hence, the positions near the surface are at approximately the same distance r from the centre of the Earth. Since the gravitational field strength g at a point is inversely proportional to the square of its distance r from the centre of the Earth ($g = \frac{GM_{\text{Earth}}}{r^2}$), the gravitational field strength is approximately constant. Hence, objects near the surface of the Earth experience a constant gravitational force. By Newton's second law, the acceleration of free fall is therefore constant.
	Note: The symbols quoted in the expression $g = \frac{GM_{\text{Earth}}}{r^2}$ must be explained.
(b) (i)	The distance fallen is less than 1.0 m which is small. Hence, the speed of the ball is low throughout the experiment. Since air resistance is proportional to the square of the speed, the air resistance acting on the ball is small at low speeds and may be neglected.
	Note: Most candidates incorrectly suggested the small surface area giving rise to low air resistance.
(ii)	1. Using $s_y = u_y t + \frac{1}{2} a_y t^2$ $t = \sqrt{\frac{2s_y}{a_y}}$ $= \sqrt{\frac{2 \times 0.441}{9.81}}$ $= 0.2998$ $\approx 0.300 \text{ s (3 s.f.)}$ 2. Distance fallen from rest to the instant the photo is taken $= 0.566 - 0.012 = 0.554 \text{ m}$ Time taken, $t' = \sqrt{\frac{2s_y}{a_y}}$ $= \sqrt{\frac{2 \times 0.554}{9.81}}$ $= 0.3361 \text{ s}$

	Time interval T $T = t' - t$ $= 0.3361 - 0.2998$ $= 0.0363 \text{ s}$
(c)	Percentage uncertainty in the stated value of T $= \frac{0.0363 - 0.033}{0.033} \times 100$ $= 10 \%$
(d) (i)	When the scale is not vertical, the 44.1 cm mark on the scale is at a decreased vertical distance of $44.1 \cos \theta$ below the point of release, where θ is the angle between the scale and the vertical. Hence, the time for the ball to fall to the 44.1 cm mark on the scale will be shorter than the time determined in (b)(ii)1.
(ii)	If the ball retains some magnetic effect, it will attract the soft iron core of the electromagnet. Hence, it will experience a smaller initial resultant downward force, resulting in a lower initial resultant downward acceleration than that of free fall g. Therefore, the ball will have a lower average downward speed and will take longer time to fall to the 44.1 cm mark on the scale than that determined in (b)(ii)1.
	Note: Most candidates did not make it clear that only the initial acceleration would be less than g.
(e)	Air resistance, which opposes the downward motion of the falling ball, increases with speed. With the ball's weight remaining constant through the fall, the resultant downward force $F_{\text{resultant}}$ acting on the ball decreases as the speed increases. According to Newton's second law, the resultant downward acceleration of the ball, $a_{\text{resultant}} = \frac{F_{\text{resultant}}}{m}$ decreases as the $F_{\text{resultant}}$ decreases. When the air resistance increases to a magnitude which is equal but opposite to the weight of the ball, the resultant downward force acting on the ball and hence its resultant downward acceleration becomes zero. Hence, it falls at a constant terminal velocity.
7 (a) (i)	$V = \frac{V_0}{2}$